Assignment

1. Solve $x - \log x - 1.2 = 0$ by regula falsi method.

Ans. $f(x) = x \log x - 1.2$

$f(1) = -1.2$

$f(2) = x \log_{10} 2 - 1.2 = 2(0.03010) - 1.2 = -0.5979 < 0$

$f(3) = 3 \log_{10} 3 - 1.2 = 3(0.4771) - 1.2 = 0.2313 > 0$

$a = 2, \quad f(a) = -0.5979.$

$b = 3, \quad f(b) = 0.2313$

$$x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)} = \frac{2(0.2313) - 3(-0.5979)}{0.2313 - (-0.5979)}$$

$= 2.7210.$

$f(x_1) = 2.7210 \log_{10} 2.7210 - 1.2 = -0.0171$

Let $a = 2.7210, \quad b = 3.$

$$x_2 = \frac{a f(b) - b f(a)}{f(b) - f(a)} = \frac{2.7210(0.2313) - 3(-0.0171)}{0.2313 - (-0.0171)}$$

$= 2.7406$

$f(x_2) = -0.00004.$

Let $a = 2.7406, \quad b = 3$

$$x_3 = \frac{2.7406(0.2313) - 3(-0.00004)}{0.2313 - (-0.00004)} = 2.7406$$

$f(x_3) = 0.00004.$

$\therefore$ solution is 2.7406

2. Solve $x - \cos x = 0$ by false position method.

Ans. Let ~~a = 0.7~~ $f(x) = x - \cos x$.

~~$f(0) = 0 - \cos 0$~~ → Let $a = 0.7$

$f(0.7) = 0.7 - \cos 0.7 = -0.0648$

$b = 1, \quad f(b) = f(1) = 0.4597$

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{0.7(0.4597) - 1(-0.0648)}{0.4597 - (-0.0648)}$$

$= 0.73706$

$f(x_1) = 0.73706 - \cos(0.73706) = -0.00338$

$a = 0.73706, \quad f(a) = -0.00338$

$b = 1, \quad f(b) = 0.4597$

$$x_2 = \frac{0.73706(0.4597) - 1(-0.00338)}{0.4597 - (-0.00338)} = 0.73897$$

≈ 0.738

3. Define the system of equation, write condition for convergence of system of equation in Gauss-siedel method.

Ans. If the equations are:

$a_1x + b_1y + c_1z = d_1 \Rightarrow x = \frac{1}{a_1}[d_1 - b_1y - c_1z]$

$a_2x + b_2y + c_2z = d_2 \Rightarrow y = \frac{1}{b_2}[d_2 - a_2x - c_2z]$

$a_3x + b_3y + c_3z = d_3 \quad z = \frac{1}{c_2}[d_3 - a_3x - b_3y]$

Then, the condition for convergence →

$|a_1| > |b_1| \not+ |c_1|$

$|b_2| > |a_2| + |c_2|$

$|c_3| > |a_3| + |b_3|$

4. Solve the following system of equations

$$28x + 4y - z = 32$$

$$x + 3y + 10z = 24$$

$$2x + 17y + 4z = 35$$

Using Gauss-Siedel method.

Ans. checking the condition for convergence;

$$|a_1| > |b_1| + |c_1| \Rightarrow |28| > |4| + |-1|$$

$$|b_2| > |a_2| + |c_2| \Rightarrow |3| \ngtr |1| + |10|$$

$$|c_3| > |a_3| + |b_3| \Rightarrow |4| \ngtr |2| + |17|$$

$\because$ the diagonal elements in the coefficient matrix are not dominant, we arrange the equation as;

$$28x + 4y - z = 32$$

$$2x + 17y + 4z = 35$$

$$x + 3y + 10 = 24$$

$$\therefore x = \frac{1}{a_1}\left[d_1 - b_1 y - c_1 z\right] = \frac{1}{28}\left[32 - 4y + z\right]$$

$$y = \frac{1}{b_2}\left[d_2 - a_2 x - c_2 z\right] = \frac{1}{17}\left[35 - 2x - 4z\right]$$

$$z = \frac{1}{c_3}\left[d_3 - a_3 x - b_3 y\right] = \frac{1}{10}\left[24 - x - 3y\right]$$

Let $y = 0$, $z = 0$.

$\therefore x' = \frac{32}{28} = 1.1429$; Let $z = 0$, $x = 1.1429$

$$y' = \frac{1}{17}\left[35 - 2 \times 1.1429 - 0\right] = \underline{1.9244}$$

Let $x = 1.1429$, $y = 1.9244$.

$$z = \frac{1}{10}\left[24 - 1.1429 - 3 \times 1.9244\right] = \underline{1.8084}.$$

Second iteration,

$$x^2 = \frac{1}{28}\left[32 - 4(1.9244) + 1.8084\right] = 0.9325.$$

$$y^2 = \frac{1}{17}\left[35 - 2(0.9325) + 4(1.8084)\right] = 1.5236$$

$$z^2 = \frac{1}{10}\left[24 - 0.9913 - 3(1.5070)\right] = 1.8488$$

3rd iteration;

$$x^3 = \frac{1}{28}\left[32 - 4(1.5236) + 1.8497\right] = 0.9913$$

$$y^3 = \frac{1}{17}\left[35 - 2(0.9913) - 4(1.8497)\right] = 1.5070$$

$$z^3 = \frac{1}{10}\left[24 - 0.9913 - 3(1.5070)\right] = 1.8488$$

Fourth iteration;

$$x^4 = \frac{1}{28}\left[32 - 4(0.15070) + 1.8488\right] = 0.9936$$

$$y^4 = \frac{1}{17}\left[35 - 2(0.9936) - 4(1.8488)\right] = 1.5069$$

$$z^4 = \frac{1}{10}\left[24 - 0.9936 - 3(1.5069)\right] = 1.8486$$

5th iteration,

$$x^5 = \frac{1}{28}\left[32 - 4(1.5069) + 1.8486\right] = 0.9936$$

$$y^5 = \frac{1}{17}\left[35 - 2(0.9936) - 4(1.8486)\right] = 1.5069.$$

$$z^5 = \frac{1}{10}\left[24 - 0.9936 - 3(1.5069)\right] = 1.8486$$

$\therefore$ the solution is;

$x = 0.9936,\quad y = 1.5069,\quad z = 1.8486.$

5. Prove that, $E^{-1/2} = \mu - \frac{1}{2}\delta$

Ans. LHS $\Rightarrow E^{-1/2}$

$$\text{RHS} \Rightarrow \mu - \frac{1}{2}\delta = \frac{E^{1/2} + E^{-1/2}}{2} - \frac{1}{2}\, E^{1/2} - E^{-1/2}$$

$$= \frac{E^{1/2} + E^{-1/2}}{2} - \frac{E^{1/2} - E^{-1/2}}{2}$$

$$= \frac{E^{1/2} + E^{-1/2} - E^{1/2} + E^{-1/2}}{2}$$

$$= \frac{2E^{-1/2}}{2} = E^{-1/2}$$

6. Prove that, $\Delta = \frac{1}{2}\delta^2 + \delta\sqrt{1+\frac{\delta^2}{4}}$

Ans. LHS ⇒ Δ.

RHS ⇒ $\frac{1}{2}\delta^2 + \delta\sqrt{1+\frac{\delta^2}{4}}$

$$= \frac{1}{2}\delta\left[\delta + 2\sqrt{1+\frac{\delta^2}{4}}\right]$$

$$= \frac{1}{2}\delta\left[\delta + \sqrt{4+\delta^2}\right]$$

$$= \frac{1}{2}\delta\left[\left(E^{1/2} - E^{-1/2}\right) + \sqrt{4+\left(E^{1/2} - E^{-1/2}\right)^2}\right]$$

$$= \frac{1}{2}\delta\left[\left(E^{1/2} - E^{-1/2}\right) + \sqrt{\left(E^{1/2} + E^{-1/2}\right)^2}\right]$$

$$= \frac{1}{2}\left(E^{1/2} - E^{-1/2}\right)\left[E^{1/2} - E^{-1/2} + E^{1/2} + E^{-1/2}\right]$$

$$= \frac{1}{2} \times 2\left[E^{1/2} - E^{-1/2}\right]E^{1/2}$$

$$= E - 1 = \Delta.$$

7. Find $y(20)$ from the following table using divided difference.

x	y	Δ	Δ^2	Δ^3	Δ^4
12	146				
		115			
18	836		454.625		
		4661.25		-218.493	
22	19481		-2167.291		21.93
		-8342.5		220.145	
24	2796		914.75		
		805			
32	9236				

Let $x_0 = 12$, $x = 20$.

$$y(x) = y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)[x_0, x_1, x_2] + \dots$$

$$\therefore y(20) = 146 + (20-12)\,115 + (20-12)(20-18)\,454.625 +$$

$$(20-12)(20-18)(20-22) \times -218.493$$

$+ (20-12)(20-18)(20-22)(20-24)(21.93)$

$y(20) = 146 + 920 + 7274 + 6991.776 + 2807.04$

$= \underline{\underline{18138.816}}$

8. The population of a certain town is given below. Find the rate of growth of population in 1931, 1941, & 1971.

year. (x)	Population in 1000 (y).	Δy.	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1931	40.62				
		20.18			
1941	60.8		1.03		
		19.15		5.49	
1951	79.95		4.46		-4.47
		23.61		1.02	
1961	103.56		5.48		
		29.09			
1971	132.65				

Ans. Let $x_0 = 1931$, $P = \frac{x-x_0}{h} = \frac{1931-1931}{10} = \frac{0}{10} = \underline{\underline{0}}$

$x = 1931$

$h = 10$ (by derivative of forward difference)

$\therefore \frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \frac{1}{4}\Delta^4 y_0 + \dots\right]$

$\left(\frac{dy}{dx}\right)_{x=1931} = \frac{1}{10}\left[20.18 - \frac{1}{2}(-1.03) + \frac{5.49}{3} - \frac{(-4.47)}{4}\right]$

$= \frac{1}{10}\left[20.18 + 0.515 + 1.83 + 1.1175\right]$

$= \underline{\underline{2.36425}}$ → Population growth in 1931

Let $x = 1941$, $x_0 = 1931$, $h = 10$.

$P = \frac{x-x_0}{10} = \frac{1941-1931}{10} = \frac{10}{10} = 1$

$\frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 + \frac{2p-1}{2!}\Delta^2 y_0 + \frac{3p^2-6p+2}{3!}\Delta^3 y_0 + \dots\right]$

$\left[\frac{dy}{dx}\right]_{x=1941} = \frac{1}{10}\left[20.18 + \frac{-1.03}{2} - \frac{5.49}{6} + \frac{(-4.47)}{12}\right]$

$= \frac{1}{10}\left[20.18 - 0.515 - 0.915 - 0.3725\right]$

$$\therefore \left(\frac{dy}{dx}\right)_{x=1941} = \underline{\underline{1.83775}} \rightarrow \text{Population growth in 1941}$$

Let $x = 1971$, $x_n = 1971$, (by using derivative of backward difference).

$h = 10$

$$P = \frac{x - x_n}{h} = \frac{1971 - 1971}{10} = \underline{\underline{0}}$$

$$\therefore \frac{dy}{dx} = \frac{1}{h}\left[\nabla y_n + \frac{\nabla^2 y_n}{2} + \frac{\nabla^3 y_n}{3} + \frac{\nabla^4 y_n}{4} + \cdots\right]$$

$$\left(\frac{dy}{dx}\right)_{x=1971} = \frac{1}{10}\left[29.09 + \frac{5.48}{2} + \frac{1.02}{3} + \frac{(-4.47)}{4}\right]$$

$$= \frac{1}{10} \times 31.05025$$

$$= \underline{\underline{3.105025}}$$

9. Find the derivative of the function at $x = 900$ from the following table.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$\Delta^6 y$
0	135						
		14					
300	149		-6				
		8		24			
600	157		18		-50		
		26		-26		70	
900	183		-8		20		-86
		18		-6		-16	
1200	201		-14		4		
		4		-2			
1500	205		-16				
		-12					
1800	193						

Let $x = x_0 = 900$, $h = 300$. $\therefore P = \frac{900 - 900}{300} = \underline{\underline{0}}$

$$\therefore \left(\frac{dy}{dx}\right)_{x=900} = \frac{1}{h}\left[\frac{1}{2}(\Delta y_0 + \Delta y_{-1}) - \frac{1}{12}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) + \cdots\right]$$

$$= \frac{1}{300}\left[\frac{1}{2}(18+26) - \frac{1}{12}(-6-26) + \frac{1}{60}(70-16)\right]$$

$$= \frac{1}{300}\left[22 + 2.6666 + 0.9\right]$$

$$= \underline{0.085222}$$

Module - II.

10. $\int_0^6 \frac{dx}{1+x}$ using trapezoidal, simpsons $1/3$, simpsons $3/8$.

Let $h = 1$.

Ans. Number of intervals $= 6$.

x :	0	1	2	3	4	5	6
$y = \frac{1}{1+x}$:	1	0.5	$1/3$	$1/4$	$1/5$	$1/6$	$1/7$

Using trapezoidal rule;

$$\int_{x_0}^{x_n} f(x)\,dx = \frac{h}{2}\left[y_0 + y_n + 2(y_1 + y_2 + y_3 + \dots + y_{n-1})\right]$$

$$\therefore \int_0^6 \frac{1}{1+x}\,dx = \frac{1}{2}\left(1 + 1/7 + 2\left(0.5 + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right)\right)$$

$$= \underline{2.0214}$$

By simpson's one-third;

$$\int_{x_0}^{x_n} f(x)\,dx = \frac{h}{3}\left[y_0 + y_n + 2(y_2 + y_4 + \dots y_{n-2}) + 4(y_1 + y_3 + \dots + y_{n-1})\right]$$

$$= \frac{1}{3}\left[1 + \frac{1}{7} + 2\left[\frac{1}{3} + \frac{1}{5}\right) + 4\left[\frac{1}{2} + \frac{1}{4} + \frac{1}{6}\right]\right)$$

$$= \frac{1}{3}\left[1 + \frac{1}{7} + \frac{16}{15} + \frac{22}{6}\right]$$

$$= \underline{1.9587}$$

By Simpson's three - eighth,

$$\int_{x_0}^{x_n} f(x)\, dx = \frac{3h}{8}\left[y_0 + y_n + 3\left(y_1 + y_2 + y_4 + \dots + y_{n-1}\right) + 2\left(y_3 + y_6 + \dots + y_{n-3}\right)\right]$$

$$= \frac{3 \times 1}{8}\left[1 + \tfrac{1}{7} + 3\left(0.5 + \tfrac{1}{3} + \tfrac{1}{5} + \tfrac{1}{6}\right) + 2\left(\tfrac{1}{4}\right)\right]$$

$$= 1.96607.$$

11. Solve $\frac{dy}{dx} = x^2 - y$, $y(0) = 1$ at $x = 0.1$ & $x = 0.2$ using Euler's method.

Ans. $f(x,y) = x^2 - y$, Let $x_0 = 0$, $y_0 = 1$ ~~h = 0.1~~ $h = 0.1$

$$y_{n+1} = y_n + hf(x_n, y_n)$$

$$y_1 = y_0 + hf(x_0, y_0)$$

$$= 1 + 0.1\, f(0, 1)$$

$$= 1 + 0.1\left(0^2 - 1\right) = 1 - 0.1$$

$$y(0.1) = 0.9$$

$$y_2 = y_1 + hf(x_1, y_1)$$

$$= 0.9 + 0.1\, f(0.1, 0.9)$$

$$= 0.9 + 0.1\left(0.1^2 - 0.9\right)$$

$$y(0.2) = 0.811$$

12. Solve $\frac{dy}{dx} = x^2 + y^2$, $y(0) = 1$ at $x = 0.1$ & $x = 0.2$ using modified Euler's theorem.

Ans. $f(x,y) = x^2 + y^2$, $y_0 = 1$, $x_0 = 0$, $h = 0.1$

$$y_{n+1} = y_n + hf\left(x_n + \frac{h}{2}, y_n + \frac{h}{2} f(x_n, y_n)\right)$$

$$y_1 = y_0 + hf\left[x_0 + \frac{h}{2}, y_0 + \frac{h}{2} \cdot f(x_0, y_0)\right]$$

$$= 1 + 0.1 f\left[0 + \frac{0.1}{2}, 1 + \frac{0.1}{2} f(0, 1)\right]$$

$$= 1 + 0.1 f[0.05, 1.05]$$

$$= 1 + 0.1 \left[0.05^2 + 0.05^2\right]$$

$y(0.1) =$ ~~1.0005~~ $\underline{\underline{1.1105}}$

$$y_2 = y_1 + hf\left[x_1 + \frac{h}{2}, y_1 + \frac{h}{2} \cdot f(x_1, y_1)\right]$$

$$= 1.1105 + 0.1 f\left[0.1 + \frac{0.1}{2}, 1.1105 + \frac{0.1}{2} f(0.1, 1.1105)\right]$$

$$= 1.1105 + 0.1\left[0.15^2 + 1.1726^2\right]$$

$y(0.2) = \underline{\underline{1.25026}}$

13. Solve $\frac{dy}{dx} = x^2y - 1$, $y(0) = 1$ at $x = 0.1$ using Taylor series.

Ans. $f(x,y) = x^2y - 1$, $y_0 = 1$, $x_0 = 0$, $h = 0.1$

By Taylor series;

$$y_{n+1} = y_n + hy_n' + \frac{h^2}{2!} y_n'' + \frac{h^3}{3!} y_n''' + \dots$$

$$\therefore y_1 = y_0 + hy_0' + \frac{h^2}{2!} y_0'' + \dots$$

Given, $y_0' = x_0^2 y_0 - 1$

$$\therefore y_0' = 0^2 \times 1 - 1 = \underline{\underline{-1}}$$

$$y'' = 2xy + x^2y'$$

$$y_0'' = 2(0 \times 1) + (0^2 \times -1) = 0.$$

$$y''' = 2y + 2xy' + 2xy' + x^2y''$$

$$y_1''' = 0 + (-1)2\times 0 + 1\times 2 + 0 = 2$$

$$\cancel{y^{iv}} = \cancel{2y + 2y' + 2xy'' + 2y' + 2xy'' + 2xy'' + x^2y'''}$$

$$= \cancel{x^2y''' + 6xy'' + 6y' + 2}$$

$$\cancel{y_1^{iv} = 0 + 0 + 4\times -1 + 2 = 2}$$

$$y^{iv} = x^2y''' + y''2x + 4xy'' + 4y' + 2y'$$

$$y_1^{iv} = 0 + 0 + 0 + (-1)4 + 2(-1)$$

$$= \underline{-6}$$

$$y(x) = y(0.1) = 1 - 0.1 + 0 + \frac{(0.1)^3}{6}\times 2 - \frac{(0.1)^4}{24}\times 6$$

$$= \underline{\underline{0.90031}}$$

14. Solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$, $y(0) = 1$ at $x = 0.2$ using R-k method of 4th order.

Ans. $$y' = \frac{y^2 - x^2}{y^2 + x^2}$$

$$x_0 = 0, \quad y_0 = 1, \quad h = 0.2$$

$$f(x_0, y_0) = f(0, 1) = \frac{1 - 0}{1 + 0} = \underline{1}$$

$$k_1 = hf(x_0, y_0) = (0.2)\times 1 = \underline{\underline{0.2}}$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.2 f(0.1, 1.1)$$

$$= 0.2\left[\frac{1.1^2 - 0.1^2}{1.1^2 + 0.1^2}\right] = 0.2\left[\frac{1.21 - 0.01}{1.21 + 0.01}\right]$$

$$= \underline{\underline{0.1967}}$$

$$k_3 = hf\left[x_0 + \frac{h}{2},\ y_0 + \frac{k_2}{2}\right]$$

$$= 0.2\, f\left[0.1,\ 1 + \frac{0.1967}{2}\right]$$

$$= 0.2 \left(\frac{1.0983^2 - 0.1^2}{1.0983^2 + 0.1^2}\right)$$

$$= \underline{\underline{0.1967.}}$$

$$k_4 = hf\left(x_0 + h,\ y_0 + k_3\right)$$

$$= 0.2\, f\left(0.2,\ 1.1967\right)$$

$$= 0.2 \left(\frac{1.1967^2 - 0.2^2}{1.1967^2 + 0.2^2}\right)$$

$$= \underline{\underline{0.1891}}$$

$$\therefore\ \Delta y = \frac{1}{6}\left[k_1 + 2k_2 + 2k_3 + k_4\right]$$

$$= \frac{1}{6}\left[0.2 + 2(0.1967) + 2(0.1967) + 0.1891\right]$$

$$y(0.2 = \underline{\underline{0.1959.}}$$